

Manual of Steel Construction

AISC Steel Construction Manual, 15th Edition

Part 1

Dimensions and Properties

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Wide-flange (W) Shapes

- Most widely used section
- Two flanges held apart by a web
- Essentially parallel inner and outer flange surfaces

Section designation

W24x55

Weight per foot

Nominal depth

Major (strong) axis

Flange

Web

Minor (weak) axis

Table 1-1.
W-Shapes
Dimensions

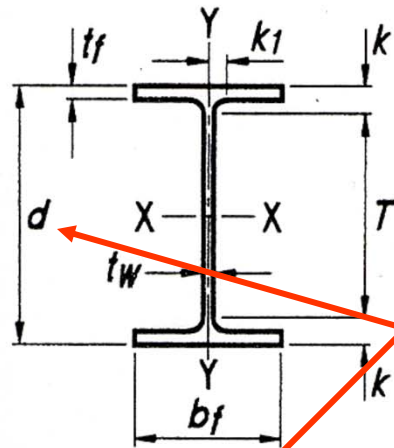
Table 1-1.
W-Shapes
Dimensions

Minor (weak) axis

Shape	Area, A in. ²	Depth, d in.		Web		Flange				Distance						
				Thickness, t_w in.		$\frac{t_w}{2}$ in.		Width, b_f in.		Thickness, t_f in.		k in.		k_1 in.	T in.	Work- able Gage [†] in.
W44×335*	98.3	44.0	44	1.02	1	$1\frac{1}{2}$	16.0	16	1.77	$1\frac{3}{4}$	2.56	$2\frac{5}{8}$	$1\frac{5}{16}$	$38\frac{3}{4}$	$5\frac{1}{2}$	
×290	85.8	43.6	$43\frac{5}{8}$	0.870	$\frac{7}{8}$	$\frac{7}{16}$	15.8	$15\frac{7}{8}$	1.58	$1\frac{9}{16}$	2.37	$2\frac{7}{16}$	$1\frac{1}{4}$	↓	↓	
×262	77.2	43.3	$43\frac{1}{4}$	0.790	$\frac{13}{16}$	$\frac{7}{16}$	15.8	$15\frac{3}{4}$	1.42	$1\frac{7}{16}$	2.21	$2\frac{1}{4}$	$1\frac{3}{16}$	↓	↓	
×230	67.7	42.9	$42\frac{7}{8}$	0.710	$\frac{11}{16}$	$\frac{3}{8}$	15.8	$15\frac{3}{4}$	1.22	$1\frac{1}{4}$	2.01	$2\frac{1}{16}$	$1\frac{3}{16}$	↓	↓	

Designation

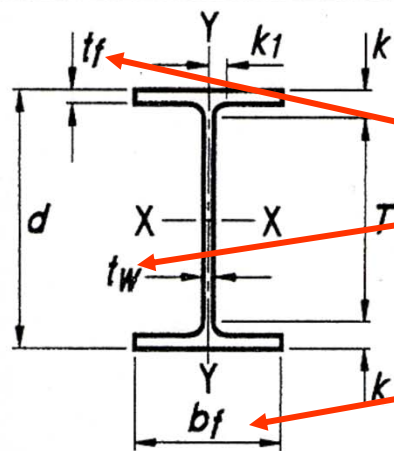
**Table 1-1.
W-Shapes
Dimensions**



Cross-sectional area

Actual depth

Shape	Area, A		Depth, d		Web			Flange			Distance					Workable Gage†
					Thickness, t _w		t _w /2	Width, b _f		Thickness, t _f	k		k ₁	T		
	in. ²	in.	in.	in.	in.	in.	in.	in.	in.	in.	in.	in.	in.	in.		
W44×335*	98.3	44.0	44	1.02	1	1/2	16.0	16	1.77	1 3/4	2.56	2 5/8	1 5/16	38 3/4	5 1/2	
×290	85.8	43.6	43 5/8	0.870	7/8	7/16	15.8	15 7/8	1.58	1 9/16	2.37	2 7/16	1 1/4	↓	↓	
×262	77.2	43.3	43 1/4	0.790	13/16	7/16	15.8	15 3/4	1.42	1 7/16	2.21	2 1/4	1 3/16	↓	↓	
×230	67.7	42.9	42 7/8	0.710	11/16	3/8	15.8	15 3/4	1.22	1 1/4	2.01	2 1/16	1 3/16	↓	↓	



**Table 1-1.
W-Shapes
Dimensions**

Web thickness

Flange properties

Shape	Area, A	Depth, d		Web		Flange				Distance					
				Thickness, t _w		t _w / 2	Width, b _f		Thickness, t _f		k		k ₁	T	Work- able Gage†
				in.	in.	in.	in.	in.	in.	in.	in.	in.	in.	in.	
W44×335*	98.3	44.0	44	1.02	1	1/2	16.0	16	1.77	1 3/4	2.56	2 5/8	1 5/16	38 3/4	5 1/2
×290	85.8	43.6	43 5/8	0.870	7/8	7/16	15.8	15 7/8	1.58	1 9/16	2.37	2 7/16	1 1/4	↓	↓
×262	77.2	43.3	43 1/4	0.790	13/16	7/16	15.8	15 3/4	1.42	1 7/16	2.21	2 1/4	1 3/16	↓	↓
×230	67.7	42.9	42 7/8	0.710	11/16	3/8	15.8	15 3/4	1.22	1 1/4	2.01	2 1/16	1 3/16	↓	↓

**Table 1-1.
W-Shapes
Dimensions**

End of fillet transition
between web and flange
Flat portion of web

Shape	Area, A in. ²	Depth, d in.	Web				Flange				Distance				Work- able Gage† in.
			Thickness, t _w in.		$\frac{t_w}{2}$ in.	Width, b _f in.		Thickness, t _f in.		k		k ₁ in.	T in.		
										in.	in.			in.	
W44×335*	98.3	44.0	44	1.02	1	$\frac{1}{2}$	16.0	16	1.77	$1\frac{3}{4}$	2.56	$2\frac{5}{8}$	$1\frac{5}{16}$	$38\frac{3}{4}$	$5\frac{1}{2}$
×290	85.8	43.6	$43\frac{5}{8}$	0.870	$\frac{7}{8}$	$\frac{7}{16}$	15.8	$15\frac{7}{8}$	1.58	$1\frac{9}{16}$	2.37	$2\frac{7}{16}$	$1\frac{1}{4}$		
×262	77.2	43.3	$43\frac{1}{4}$	0.790	$\frac{13}{16}$	$\frac{7}{16}$	15.8	$15\frac{3}{4}$	1.42	$1\frac{7}{16}$	2.21	$2\frac{1}{4}$	$1\frac{3}{16}$		
×230	67.7	42.9	$42\frac{7}{8}$	0.710	$\frac{11}{16}$	$\frac{3}{8}$	15.8	$15\frac{3}{4}$	1.22	$1\frac{1}{4}$	2.01	$2\frac{1}{16}$	$1\frac{3}{16}$		

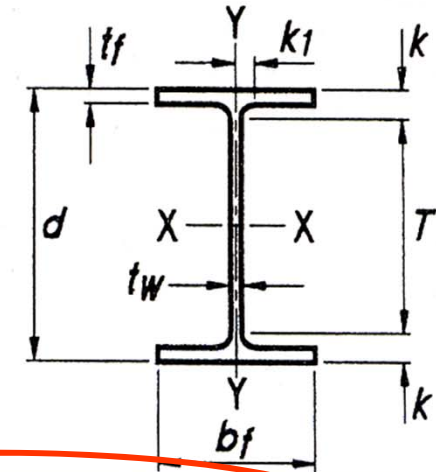
End of fillet transition
between web and flange

Flat portion of web

Spacing between rows
of bolts in flange

Second moment,
elastic section modulus,
radius of gyration,
plastic section modulus
for strong and weak axes

Table 1-1 (cont.).
W-Shapes
Properties



Weight per foot

Flange and web stability
parameters

Nom- inal Wt.	Compact Section Criteria			X_1	$X_2 \times 10^6$	Axis X-X				Axis Y-Y			
	$\frac{b_f}{2t_f}$	$\frac{h}{t_w}$	F_y''' ksi			I	S	r	Z	I	S	r	Z
lb/ft				ksi	(1/ksi) ²	in. ⁴	in. ³	in.	in. ³	in. ⁴	in. ³	in.	in. ³
335	4.51	38.1	44.3	2430	5110	31100	1410	17.8	1620	1200	151	3.50	236
290	5.01	44.7	32.2	2150	8170	27100	1240	17.8	1420	1050	132	3.49	206
262	5.55	49.2	26.6	1930	12300	24200	1120	17.7	1270	927	118	3.46	183
230	6.45	54.8	21.5	1690	21100	20800	971	17.5	1100	796	101	3.43	157

Used for beam strength
calculations

M-Shapes

- Not classified in ASTM 6 as W-, S- or HP- shapes
- Same properties (A , d , t_w , b_f , etc) as W- shapes

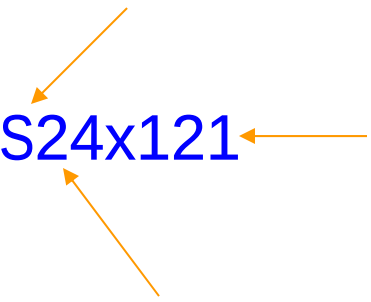
HP-Shapes

- Also known as bearing piles
- Similar to W-shapes, except their webs and flanges are of equal thickness and the depth and flange width are nominally equal for a given designation

American Standard (S) Shapes

- 16-2/3% slope on inner flange surface

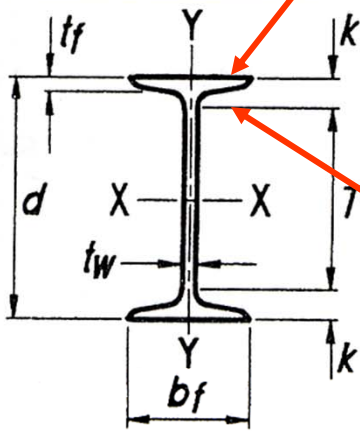
Section designation
S24x121
Weight per foot
Nominal depth



- Relatively narrow flange when compared to W shapes

Narrow flange

**Table 1-3.
S-Shapes
(American Standard Beams)
Dimensions**

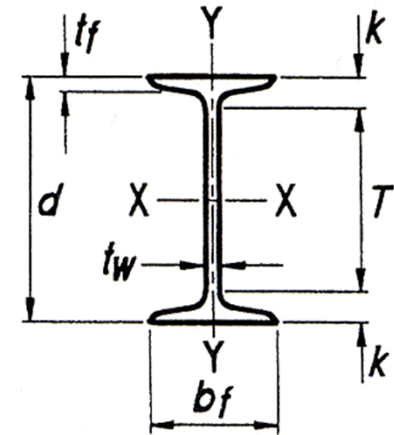


Note slope on inside of flange

Shape	Area, <i>A</i>	Depth, <i>d</i>		Web			Flange				Distance		
				Thickness, <i>t_w</i>		$\frac{t_w}{2}$	Width, <i>b_f</i>		Thickness, <i>t_f</i>		<i>k</i>	<i>T</i>	Work- able Gage†
	in. ²	in.		in.		in.	in.		in.		in.	in.	in.
S24×121	35.5	24.5	24 1/2	0.800	13/16	7/16	8.05	8	1.09	1 1/16	2	20 1/2	4
×106	31.1	24.5	24 1/2	0.620	5/8	5/16	7.87	7 7/8	1.09	1 1/16	2	20 1/2	4

Same properties as for W shapes

Table 1-3 (cont.).
S-Shapes
(American Standard Beams)
Properties



Nom- inal Wt.	Compact Section Criteria			X_1	$X_2 \times 10^6$	Axis X-X				Axis Y-Y			
	$\frac{b_f}{2t_f}$	$\frac{h}{t_w}$	F_y'''			I	S	r	Z	I	S	r	Z
	lb/ft		ksi	ksi	(1/ksi) ²	in. ⁴	in. ³	in.	in. ³	in. ⁴	in. ³	in.	in. ³
121	3.69	25.9	—	3310	1770	3160	258	9.43	306	83.0	20.6	1.53	36.3
106	3.61	33.4	57.8	2960	2470	2940	240	9.71	279	76.8	19.5	1.57	33.4

Channels

- 16-2/3% slope on inner flange surface

Section designation
C15x50
Weight per foot
Actual depth

MC – Miscellaneous channel – 2 on 12 slope on inner flange

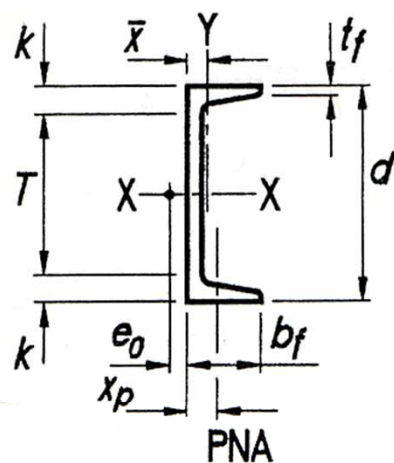


Table 1-5.
C-Shapes
(American Standard Channels)
Dimensions

Property for design

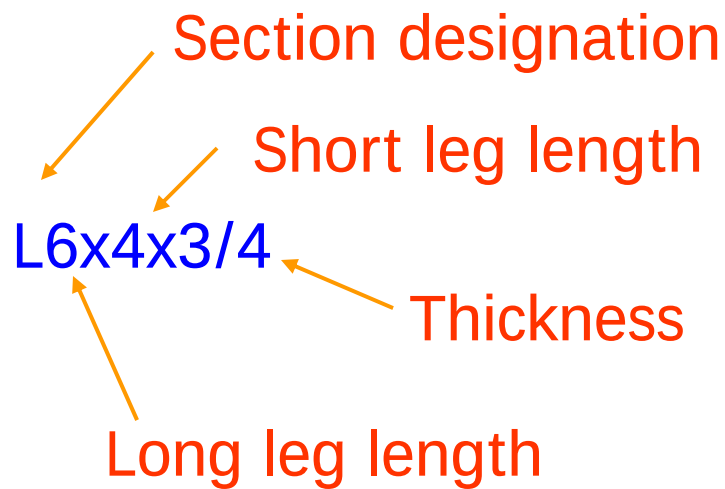
Actual depth

Property for detailing

Shape	Area, A	Depth, d		Web			Flange			Distance		
				Thickness, t_w		$\frac{t_w}{2}$	Width, b_f		Thickness, t_f	k	T	Work- able Gage [†]
	in. ²	in.	in.	in.	in.	in.	in.	in.	in.	in.	in.	in.
C15×50	14.7	15.0	15	0.716	$\frac{11}{16}$	$\frac{3}{8}$	3.72	$3\frac{3}{4}$	0.650	$1\frac{7}{16}$	$12\frac{1}{8}$	$2\frac{1}{4}$
×40	11.8	↓	↓	0.520	$\frac{1}{2}$	$\frac{1}{4}$	3.52	$3\frac{1}{2}$	↓	↓	↓	2
×33.9	9.95	↓	↓	0.400	$\frac{3}{8}$	$\frac{3}{16}$	3.40	$3\frac{3}{8}$	↓	↓	↓	2

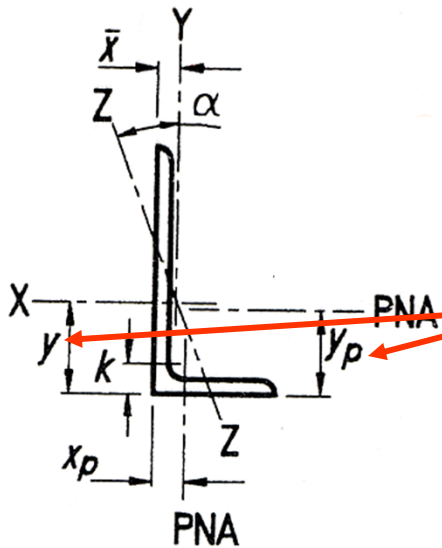
Angles

Section designation
Short leg length
L6x4x3/4
Thickness
Long leg length

A diagram of an L-shaped angle section. The section is represented by a blue shaded area. A horizontal red line is drawn across the top of the section. Four orange arrows point from text labels to the dimensions of the section: 'Section designation' points to the text 'L6x4x3/4', 'Short leg length' points to the vertical leg, 'Thickness' points to the thickness of the section, and 'Long leg length' points to the horizontal leg.

- Major axes do not correspond to X and Y axes

Table 1-7.
Angles
(L-Shapes)
Properties



Location of plastic centroid

Location of elastic centroid

X axis properties

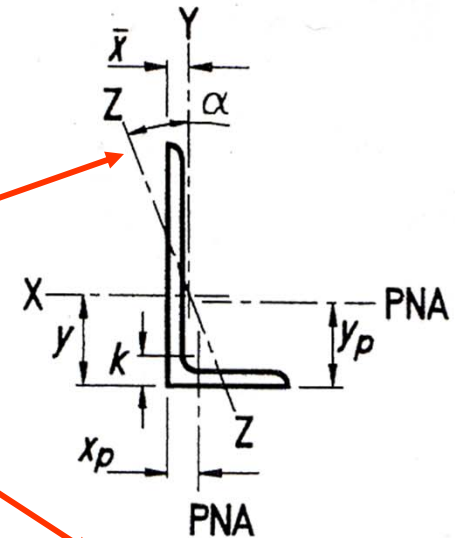
Shape	k in.	Wt. lb/ft	Area, A in. ²	Axis X-X					
				I	S	r	$\bar{y}$	Z	y_p
				in. ⁴	in. ³	in.	in.	in. ³	in.
L8×8×1 1/8	1 3/4	57.2	16.8	98.1	17.5	2.41	2.40	31.6	1.05
×1	1 5/8	51.3	15.1	89.1	15.8	2.43	2.36	28.5	0.943
×7/8	1 1/2	45.3	13.3	79.7	14.0	2.45	2.31	25.3	0.832
×3/4	1 3/8	39.2	11.5	69.9	12.2	2.46	2.26	22.0	0.720
×5/8	1 1/4	33.0	9.69	59.6	10.3	2.48	2.21	18.6	0.606
×9/16	1 3/16	29.8	8.77	54.2	9.33	2.49	2.19	16.8	0.548
×1/2	1 1/8	26.7	7.84	48.8	8.36	2.49	2.17	15.1	0.490

Designation

Table 1-7 (cont.).
Angles
(L-Shapes)
Properties

Minor (weak) axis

Y axis properties



Shape	Axis Y-Y						Axis Z-Z		Q_s^*
	I	S	r	$\bar{x}$	Z	x_p	r	Tan α	$F_y = 36$ ksi
	in. ⁴	in. ³	in.	in.	in. ³	in.	in.		
L8×8×1 1/8	98.1	17.5	2.41	2.40	31.6	1.05	1.56	1.00	—
×1	89.1	15.8	2.43	2.36	28.5	0.943	1.56	1.00	—
×7/8	79.7	14.0	2.45	2.31	25.3	0.832	1.57	1.00	—
×3/4	69.9	12.2	2.46	2.26	22.0	0.720	1.57	1.00	—
×5/8	59.6	10.3	2.48	2.21	18.6	0.606	1.58	1.00	0.997
×9/16	54.2	9.33	2.49	2.19	16.8	0.548	1.58	1.00	0.959
×1/2	48.8	8.36	2.49	2.17	15.1	0.490	1.59	1.00	0.912

Tees

- WT – cut from W shape

WT22x131 is cut from W44x262

- ST – cut from S shape
- MT – cut from M shape

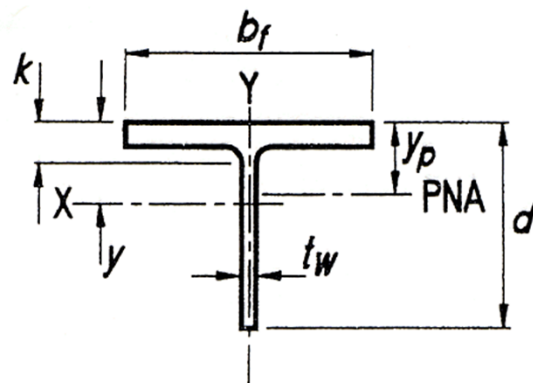
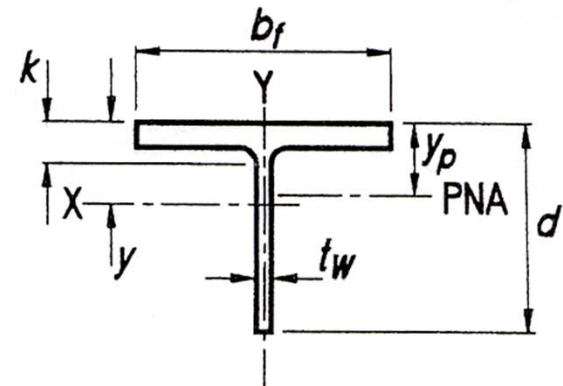


Table 1-8.
WT-Shapes
(Structural Tees Split
from W-Shapes)
Dimensions

Stem, not web

Shape	Area, A	Depth of Tee, d		Stem				Flange				Distance		
				Thickness, t _w		t _w 2	Area	Width, b _f		Thickness, t _f		k		Work- able Gage†
	in. ²	in.	in.		in.	in. ²	in.		in.		in.	in.	in.	
WT22×167.5*	49.1	22.0	22	1.02	1	1/2	22.5	16.0	16	1.77	1 3/4	2.56	2 5/8	5 1/2 ↓
×145	42.9	21.8	21 3/4	0.870	7/8	7/16	19.0	15.8	15 7/8	1.58	1 9/16	2.37	2 7/16	
×131	38.6	21.7	21 5/8	0.790	13/16	7/16	17.1	15.8	15 3/4	1.42	1 7/16	2.21	2 1/4	
×115	33.8	21.5	21 1/2	0.710	11/16	3/8	15.2	15.8	15 3/4	1.22	1 1/4	2.01	2 1/16	

Table 1-8 (cont.).
WT-Shapes
(Structural Tees Split
from W-Shapes)
Properties



Nom- inal Wt.	$\frac{h}{t_w}$	Axis X-X						Axis Y-Y				Q_s^{**} $F_y = 50 \text{ ksi}$
		I	S	r	$\bar{y}$	Z	y_p	I	S	r	Z	
lb/ft		in. ⁴	in. ³	in.	in.	in. ³	in.	in. ⁴	in. ³	in.	in. ³	
167.5	19.1	2160	131	6.63	5.51	233	1.54	600	75.3	3.50	118	0.817
145	22.3	1840	111	6.55	5.27	197	1.35	523	66.1	3.49	103	0.636
131	24.6	1650	100	6.53	5.20	177	1.23	463	58.8	3.46	91.3	0.532
115	27.4	1440	88.6	6.53	5.17	157	1.07	398	50.5	3.43	78.3	0.438

Reduction factor for
slender stiffened
compression elements

Hollow Structural Shapes (HSS)

- Rectangular (or square)
- Round

Steel Pipe

Pipe diameter (Std., X-Strong, XX-Strong)

For example, Pipe 5 Std.

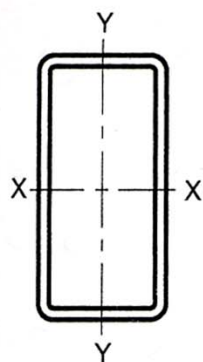
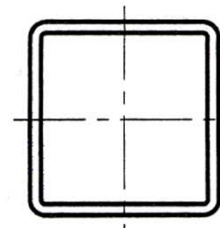


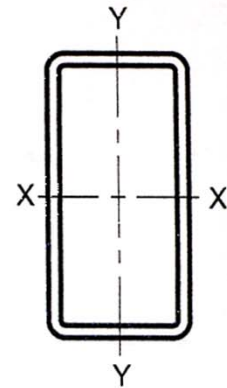
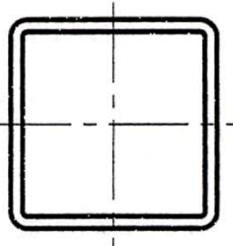
Table 1-11.
Rectangular (and Square) HSS
Dimensions and Properties



Shape	Wall Thickness, t		Nom- inal Wt. lb/ft	Area, A in. ²	$\frac{b}{t}$	$\frac{h}{t}$	Axis X-X			
	nominal	design					I	S	r	Z
	in.	in.					in. ⁴	in. ³	in.	in. ³
HSS20×12	5/8	0.581	127	35.0	17.7	31.4	1880	188	7.33	230
	1/2	0.465	103	28.3	22.8	40.0	1550	155	7.39	188
	3/8	0.349	78.4	21.5	31.4	54.3	1200	120	7.45	144
	5/16	0.291	65.8	18.1	38.2	65.7	1010	101	7.48	122

Hollow Structural Shapes (HSS)

Table 1-11 (cont.).
Rectangular (and Square) HSS
Dimensions and Properties



Torsion and
warping constants

Shape	Wall Thickness, t		Axis Y-Y				Torsion		Surface Area Per Foot ft ²
	nominal	design	I	S	r	Z	J	C	
	in.	in.	in. ⁴	in. ³	in.	in. ³	in. ⁴	in. ³	
HSS20×12	5/8	0.581	851	142	4.93	162	1890	257	5.17
	1/2	0.465	705	117	4.99	132	1540	209	5.20
	3/8	0.349	547	91.1	5.04	102	1180	160	5.23
	5/16	0.291	464	77.3	5.07	85.8	997	134	5.25

Diameter over design thickness

Nominal versus design thickness

 Table 1-12. Round HSS Dimensions and Properties												
Shape	Wall Thickness, t		Nom- inal Wt. lb/ft	Area, A in. ²	$\frac{D}{t}$	I in. ⁴	S in. ³	r in.	Z in. ³	Torsion		Surf. Area Per Ft ft ²
	nominal	design								J	C	
	in.	in.								in. ⁴	in. ³	
HSS20.000	0.500	0.465	104	28.5	43.0	1360	136	6.91	177	2720	272	5.24
	0.375	0.349	78.7	21.5	57.3	1040	104	6.95	135	2080	208	5.24

Double Angles

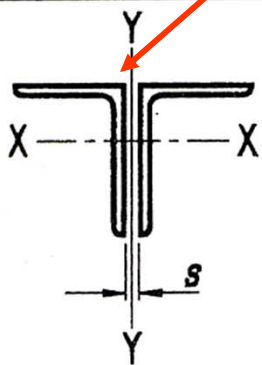
2L6x4x3/4

- Major axes are now x and y
- X axis properties may be obtained from x axis properties of single angle
- Y axis properties depend on separation between backs angles and whether LLBB or SLBB

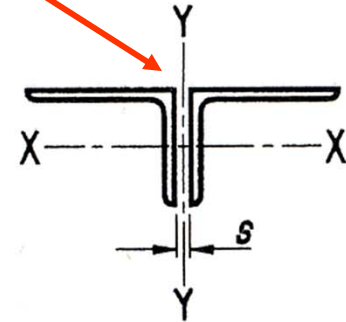
Equal leg angles

Long legs back-to-back

Short legs back-to-back



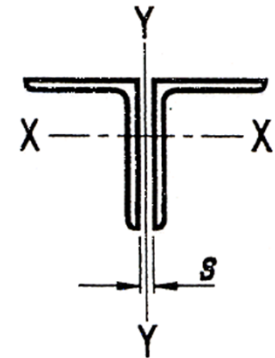
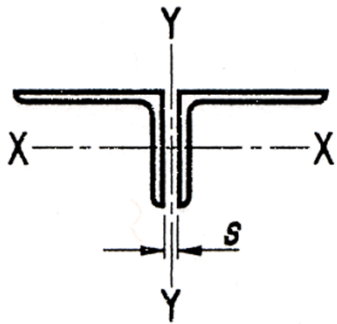
**Table 1-14.
Double Angles
(2L-Shapes)
Properties**



Shape	Axis Y-Y						Q_s^*			
	Radii of Gyration									
	LLBB			SLBB			LLBB		SLBB	
	Separation s , in.			Separation s , in.			Angles in Contact	Angles Separated	Angles in Contact	Angles Separated
	0	$3/8$	$3/4$	0	$3/8$	$3/4$				
2L8×8×1 1/8	3.41	3.54	3.68	3.41	3.54	3.68	—	—	—	—
×1	3.39	3.52	3.66	3.39	3.52	3.66	—	—	—	—
×7/8	3.36	3.50	3.63	3.36	3.50	3.63	—	—	—	—
×3/4	3.34	3.47	3.61	3.34	3.47	3.61	—	—	—	—
×5/8	3.32	3.45	3.58	3.32	3.45	3.58	—	0.997	—	0.997
×9/16	3.31	3.44	3.57	3.31	3.44	3.57	—	0.959	—	0.959
×1/2	3.30	3.43	3.56	3.30	3.43	3.56	0.998	0.912	0.998	0.912

Unequal leg angles

**Table 1-14 (cont.).
Double Angles
(2L-Shapes)
Properties**



Shape	Axis Y-Y						Q_s^*			
	Radii of Gyration									
	LLBB			SLBB			LLBB		SLBB	
	Separation s , in.			Separation s , in.			Angles in Contact	Angles Separated	Angles in Contact	Angles Separated
	0	$\frac{3}{8}$	$\frac{3}{4}$	0	$\frac{3}{8}$	$\frac{3}{4}$				
2L5×3 $\frac{1}{2}$ × $\frac{3}{4}$	1.39	1.53	1.68	2.33	2.47	2.62	—	—	—	—
× $\frac{5}{8}$	1.37	1.50	1.65	2.30	2.45	2.59	—	—	—	—
× $\frac{1}{2}$	1.35	1.48	1.62	2.28	2.42	2.57	—	—	—	—
× $\frac{3}{8}$	1.33	1.46	1.59	2.26	2.39	2.54	—	0.983	—	0.983
× $\frac{5}{16}$	1.32	1.44	1.58	2.25	2.38	2.52	—	0.912	0.998	0.912
× $\frac{1}{4}$	1.31	1.43	1.57	2.23	2.37	2.51	—	0.804	0.894	0.804

Double Channels

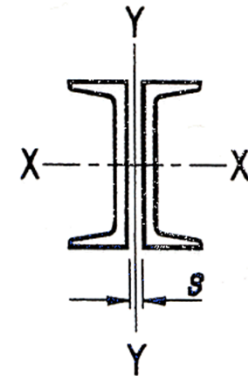
- Designated as 2C or 2MC

2C15x50

- Y axis properties depend on back-to-back separation
- X axis properties can be obtained from x axis properties of single channel

Y axis properties depend
on back-to-back
distance between
individual channels

Table 1-15.
2C-Shapes
(Double Channels)
Properties



Shape	Axis Y-Y											
	Separation s, in.											
	0				³ / ₈				³ / ₄			
	<i>I</i> in. ⁴	<i>S</i> in. ³	<i>r</i> in.	<i>Z</i> in. ³	<i>I</i> in. ⁴	<i>S</i> in. ³	<i>r</i> in.	<i>Z</i> in. ³	<i>I</i> in. ⁴	<i>S</i> in. ³	<i>r</i> in.	<i>Z</i> in. ³
2C15×50	40.7	11.0	1.18	30.7	50.5	12.9	1.31	36.2	62.4	15.3	1.46	41.7
×40	32.6	9.25	1.18	22.9	40.2	10.9	1.31	27.3	49.6	12.7	1.45	31.7
×33.9	28.5	8.38	1.20	19.0	35.1	9.78	1.33	22.7	43.1	11.4	1.47	26.4

$$\frac{VQ}{Ib}$$

**Table 1-25.
W-Shapes
Torsional Properties**



Shape	Torsional Constant, J	Warping Constant, C_w	$\sqrt{\frac{EC_w}{GJ}}$	Normalized Warping Constant, W_{no}	Warping Statical Moment, S_w	Statical Moment	
	in. ⁴	in. ⁶	in.	in. ²	in. ⁴	Q_f in. ³	Q_w in. ³
W44×335*	74.4	536000	137	168	1190	279	805
×290	51.5	464000	153	166	1040	248	704
×262	37.7	407000	167	165	922	222	630
×230	24.9	346000	190	164	789	191	546

Table 2-4 Applicable ASTM Specifications /Shapes

Table 1

Applicable ASTM Specifications for Various Structural Shapes															
Steel Type	ASTM Designation		F _y Min. Yield Stress (ksi)	F _u Tensile Stress* (ksi)	Applicable Shape Series										
					W	M	S	HP	C	MC	L	HSS		Pipe	
												Rect.	Round		
Carbon	A36		36	58–80 ^b											
	A53 Gr. B		35	60											
	A500	Gr. B	42	58											
			46	58											
		Gr. C	46	62											
			50	62											
	A501	Gr. A	36	58											
		Gr. B	50	70											
	A529 ^c	Gr. 50	50	65–100											
		Gr. 55	55	70–100											
High-Strength Low-Alloy	A572	Gr. 42	42	60											
		Gr. 50	50	65 ^d											
		Gr. 55	55	70											
		Gr. 60*	60	75											
		Gr. 65*	65	80											
	A618 ^f	Gr. I & II	50 ^g	70 ^g											
		Gr. III	50	65											
	A913	50	50 ^h	60 ^h											
		60	60	75											
		65	65	80											
		70	70	90											
	A992		50–65 ⁱ	65 ⁱ											
	Corrosion Resistant High-Strength Low-Alloy	A242	42 ^j	63 ^j											
			46 ^k	67 ^k											
50 ^l			70 ^l												
A588		50	70												
A847		50	70												

Preferred material specification.

Other applicable material specification, the availability of which should be confirmed prior to specification.

